

CIS 6270

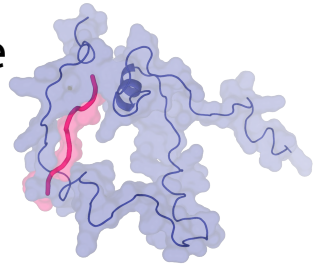
Module 2: Flow Matching

Lecture 2.1: ODEs and Continuous Probability Transport

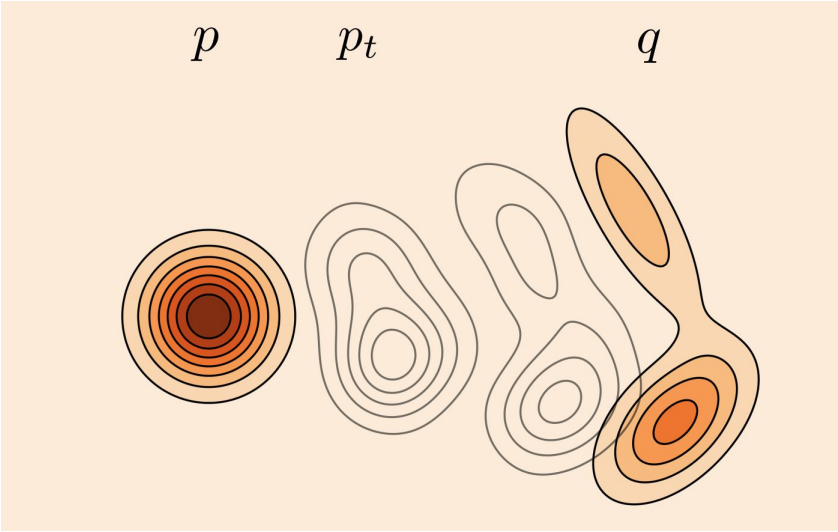
Tue, Sep 8, 2026

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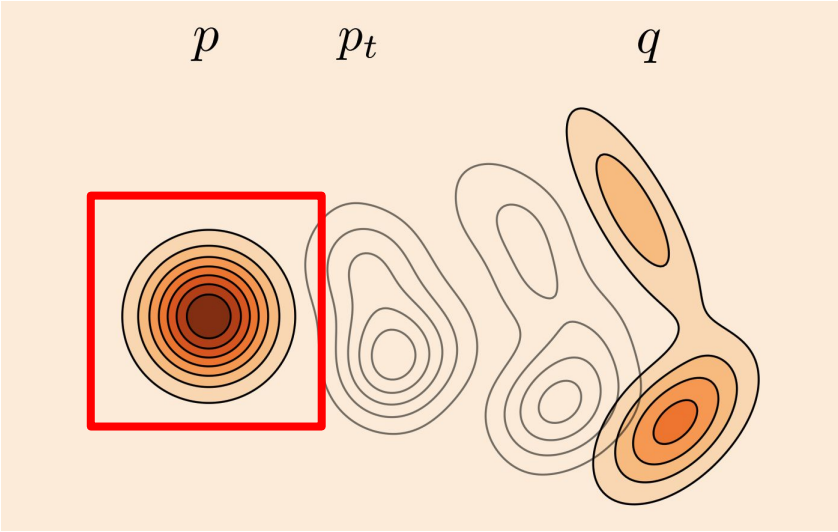


We’re still here.



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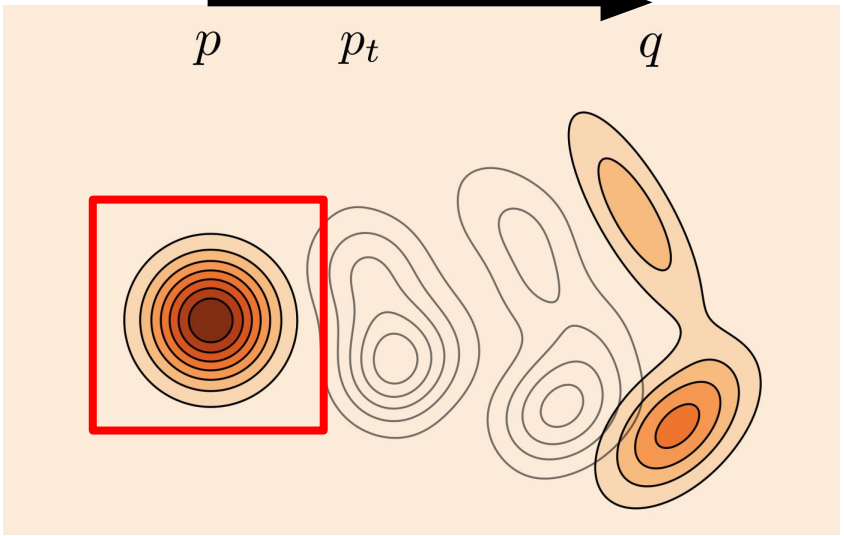
The initial random state X_0 is sampled from the initial probability distribution p_0.



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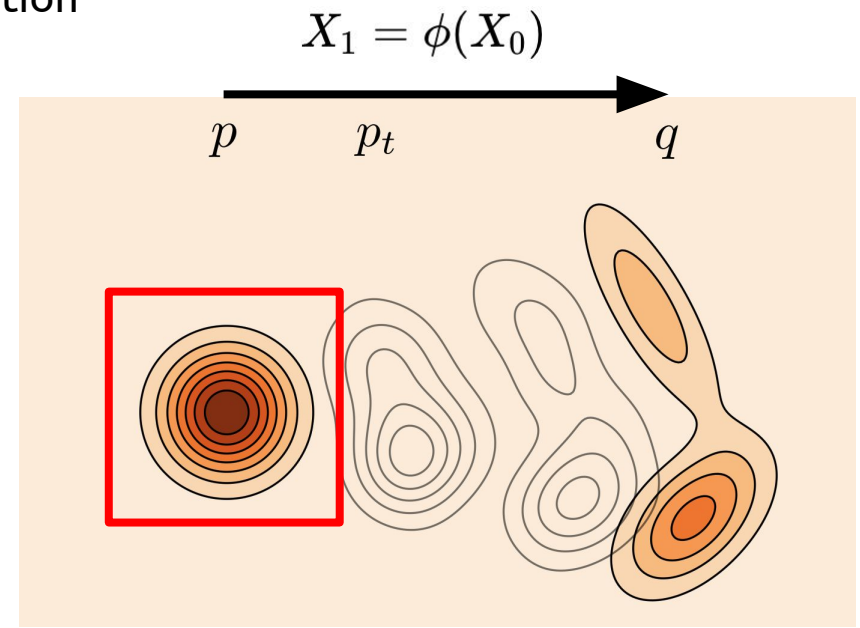
The sample then passes through a transformation.

$$X_1 = \phi(X_0)$$



We’re still here.

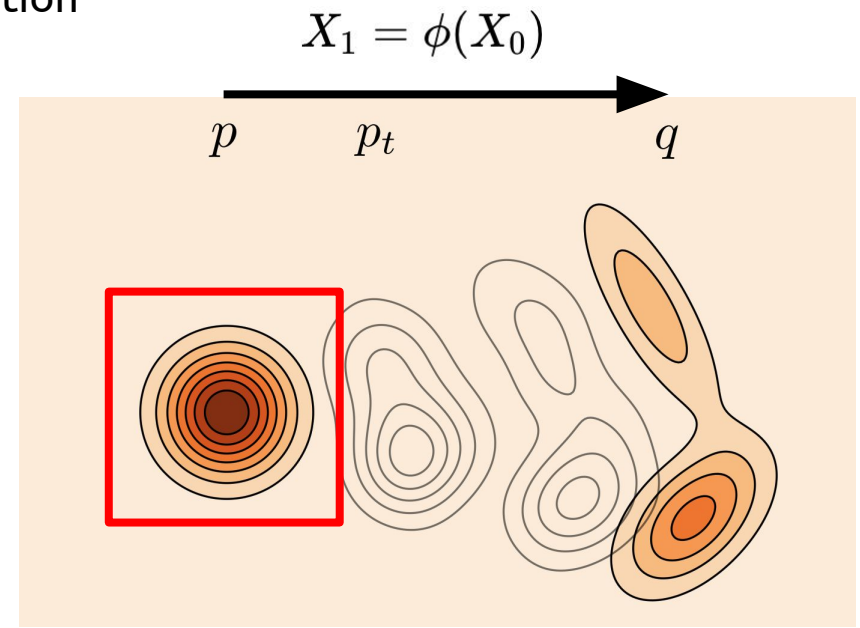
Applying the transformation to every possible initial sample transports the entire distribution



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The final probability distribution p_1 equals the pushforward of the initial probability distribution p_0 through the transformation ϕ.

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Those ideas tell us how a completed transformation changes samples and probability densities.

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Those ideas tell us how a completed transformation changes samples and probability densities.

A continuous generative model also describes everything that happens between the initial sample and the final sample.

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The family begins with the identity map: $\phi_0(x) = x$

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$$(\phi_1)_{\#}p_0 \approx p_{\text{data}}$$

How can we construct the entire family of transformations ϕ_t?

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As h approaches zero, the average velocity approaches the instantaneous velocity.

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The derivative describes the direction and speed of motion at one instant. It converts a continuous trajectory into a local motion rule.

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That local rule still needs to specify which velocity belongs to each possible state

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Once a moving sample reaches x_t, it receives the velocity assigned to that location and time.

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The ODE gives one velocity at every point along the path.

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Following these velocities through time produces the complete trajectory.

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The state moves by one-half unit during every quarter unit of time.

$$\begin{aligned} x_0 &= 1, \\ x_{0.25} &= 1.5, \\ x_{0.50} &= 2.0, \\ x_{0.75} &= 2.5, \\ x_1 &= 3.0. \end{aligned}$$

The velocity field translates the entire collection without expanding, contracting, or bending it.

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The integral form of the ODE makes this statement exact.

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We use s as an intermediate time variable because t will denote the final time.

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The state x_t equals the initial state x_0 plus the velocity $v_s(x_s)$ accumulated along the trajectory from time zero to time t.

Constant velocity produces a straight trajectory with constant speed.

$$\int_0^t \frac{dx_s}{ds}\, ds = x_t - x_0$$

Substituting this result gives:

$$x_t - x_0 = \int_0^t v_s(x_s)\, ds$$

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Those behaviors appear when velocity depends on the current state.

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We can solve this ODE exactly.

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Divide both sides by x_t and multiply by dt.

$$\frac{1}{x_t}\, dx_t = a\, dt$$

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The left side gives a difference of logarithms.

$$\log |x_t| - \log |x_0| = at$$

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Positive a produces exponential expansion. Negative a produces exponential contraction.

The solution shows how a local velocity rule creates a global transformation.

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$$x_t = 2^t x_0.$$

$$x_1 = 2^1(2) = 4.$$

$$x_t = x_0 e^{at}$$

The state doubles during one unit of time.

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Now follow three initial points.

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The distance between the outer points increases from two to four. The velocity field expands the collection around zero.

There is also exponential contraction

$$x_t = x_0 e^{at}$$

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The solution becomes

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After one unit of time, they become:

$$-\frac{1}{2}, \qquad 0, \qquad \frac{1}{2}$$

The distance between the outer points decreases from two to one. The velocity field contracts the collection toward zero.

Choose

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This leads to vector-valued ODEs.

Multidimensional velocity fields

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Let the state contain d coordinates

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The velocity field also has d components

$$v_t(x_t) = \begin{bmatrix} v_{t,1}(x_t) \\ v_{t,2}(x_t) \\ \vdots \\ v_{t,d}(x_t) \end{bmatrix}$$

The vector ODE represents d coupled scalar equations

Let the state contain d coordinates

$$x_t = \begin{bmatrix} x_{t,1} \\ x_{t,2} \\ \vdots \\ x_{t,d} \end{bmatrix} \in \mathbb{R}^d$$

$$\boxed{\frac{dx_{t,i}}{dt} = v_{t,i}(x_t), \qquad i = 1, \ldots, d}$$

The velocity field also has d components

$$v_t(x_t) = \begin{bmatrix} v_{t,1}(x_t) \\ v_{t,2}(x_t) \\ \vdots \\ v_{t,d}(x_t) \end{bmatrix}$$

Consequently, the velocity of one coordinate may depend on every coordinate.

Let the state contain d coordinates

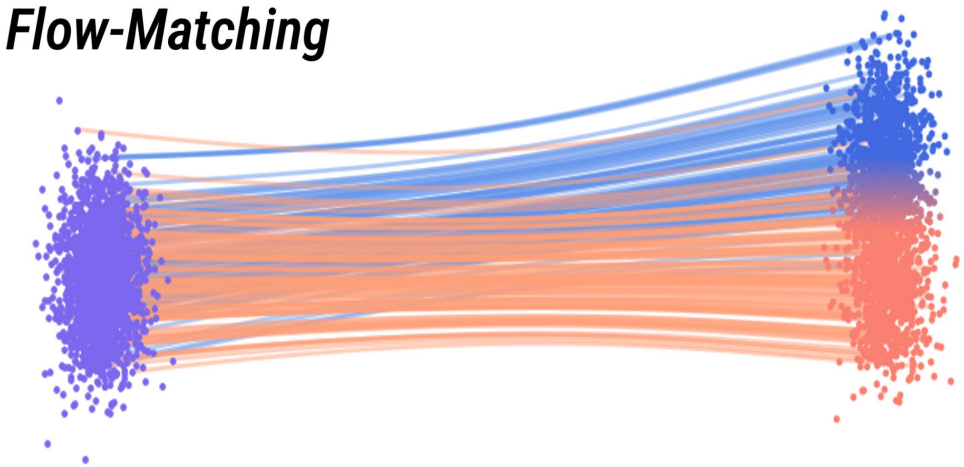
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Coupling allows trajectories to curve



Consider the velocity field

$$v(x_1, x_2) = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

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$$x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The solution is

$$x_t = \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$$

At time 0

$$v(x_1, x_2) = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

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At time π divided by two

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The trajectory moves counterclockwise around the unit circle.

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We can verify that the distance from the origin remains constant

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The solution is

$$x_t = \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$$

$$\|x_t\|^2 = \cos^2 t + \sin^2 t = 1$$

$$x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$x_{\pi/2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

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The velocity field changes direction continuously while preserving the radius.

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So a simple local rule can generate a curved global trajectory!

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Now need to connect those trajectories to the family of maps ϕ_t introduced at the beginning

Fix an initial state x_0. Solve the ODE until time t.

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The flow map gives the accumulated transformation after following that motion for a finite amount of time.

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The map ϕ_t develops according to the velocities assigned along its trajectories. Once the velocity field is known, solving the ODE constructs the transformation.

But can a single initial state always produces a single well-defined trajectory??

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Existence, uniqueness, and noncrossing trajectories

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The distance between the velocities $v_t(x)$ and $v_t(y)$ is at most the Lipschitz constant L multiplied by the distance between the states x and y.

This condition prevents the velocity from changing infinitely fast across nearby positions. Under standard assumptions, it gives a unique trajectory for each initial state.

Uniqueness also constrains the geometry.

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This property supports invertibility. If the solution remains well defined, integrating backward through time recovers the earlier state.

A learned velocity field produced by a neural network usually has no closed-form solution (simpler examples may)

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We need a numerical ODE solver!!

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Euler’s method assumes that the velocity stays approximately constant during one short step. The solver recomputes the velocity after reaching the next state.

A simple example

Consider

$$\frac{dx_t}{dt} = x_t, \qquad x_0 = 1$$

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$$x(1) = e \approx 2.7183$$

A smaller step size evaluates the velocity more frequently and usually reduces the error.

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When a neural network represents v_t, every velocity evaluation requires a neural-network evaluation

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We now have a practical procedure for moving one initial state.

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A generative model begins with a random initial state, so we need to connect these individual numerical trajectories to probability distributions.

Let the initial state be random.

$$X_0 \sim p_0$$

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Each sampled value follows the same velocity field.

$$\boxed{\frac{dX_t}{dt} = v_\theta(X_t, t)}$$

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Once X_0 has been sampled, the ODE trajectory is deterministic. Different initial samples follow different deterministic trajectories, so X_t remains a random variable.

Solving the ODE defines the flow map.

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The generative process now has a complete sample-level description.

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Applying ϕ_t to the entire initial population produces the distribution at time t.

$$\boxed{p_t = (\phi_t)_{\#} p_0}$$

First, sample X_0 from p_0.

$$X_t = \phi_t(X_0)$$

Applying ϕ_t to the entire initial population produces the distribution at time t.

$$\boxed{p_t = (\phi_t)_{\#} p_0}$$

$$X_0 \sim p_0$$

Transport it through the learned velocity field v_θ

$$X_t = \phi_t(X_0)$$

Applying ϕ_t to the entire initial population produces the distribution at time t.

$$\boxed{p_t = (\phi_t)_{\#} p_0}$$

$$X_0 \sim p_0 \quad \longrightarrow \quad \frac{dX_t}{dt} = v_\theta(X_t, t)$$

Which converts into a final random state X_1 with distribution p_1

$$X_t = \phi_t(X_0)$$

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The goal of learning is

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$$p_1 \approx p_{\text{data}}$$

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$$\boxed{\text{velocity field} \longrightarrow \text{ODE trajectories} \longrightarrow \text{flow map} \longrightarrow \text{transported distribution}}$$

Say you start with Gaussian noise

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Its solution is

$$X_t = 2^t X_0$$

Say you start with Gaussian noise

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$$X_{0.5} = \sqrt{2} X_0.$$

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$$X_t = 2^t X_0$$

Say you start with Gaussian noise

$$X_0 \sim \mathcal{N}(0, 1)$$

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The deterministic velocity field spreads the initial samples across a wider region

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Since total probability remains one, the probability density near the center must decrease as the occupied region expands

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So how does the probability density $p_t(x)$ change while all those samples move?

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The density $p_t(x)$ measures probability per unit volume near x. Integrating the density over A gives the total probability contained in that region.

Imagine drawing a fixed box around part of the state space.

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To quantify that crossing, we need probability flux.

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In one dimension, the sign of $j_t(x)$ indicates the direction of probability motion. Positive flux points toward increasing x. Negative flux points toward decreasing x.

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Conservation of probability

Consider the fixed interval:

$$A = [a, b]$$

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More probability entering than leaving causes the mass inside the interval to increase. More probability leaving than entering causes it to decrease. Equal incoming and outgoing fluxes leave the total mass unchanged.

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Using $j_t(x) = p_t(x)v_t(x)$, we obtain:

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We can also move the time derivative inside the integral.

$$\frac{d}{dt}\int_a^b p_t(x)\,dx = \int_a^b \frac{\partial p_t(x)}{\partial t}\,dx$$

Substitute both integral expressions

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This equation holds for every interval [a, b]. Therefore, the integrand must equal zero.

Density decreases where more probability flows away than arrives. Density increases where more probability arrives than leaves.

$$\boxed{\frac{\partial p_t(x)}{\partial t} + \frac{\partial}{\partial x}\left[p_t(x) v_t(x)\right] = 0}$$

This captures probability flowing through two endpoints

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Can we make it hold over EVERY point?

Local density change

The divergence theorem converts the flux across the boundary into an integral over the interior:

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- A spatially varying density can move through the region
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Expanding the divergence separates these effects.

We can expand the Continuity Equation

Use the product rule for divergence

$$\nabla \cdot [p_t(x) v_t(x)] = v_t(x) \cdot \nabla p_t(x) + p_t(x) \nabla \cdot v_t(x)$$

We can expand the Continuity Equation

Use the product rule for divergence

$$\nabla \cdot [p_t(x)v_t(x)] = v_t(x) \cdot \nabla p_t(x) + p_t(x)\nabla \cdot v_t(x)$$

Substitute this identity into the continuity equation.

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The term $p_t(x) \nabla \cdot v_t(x)$ accounts for local volume change.

When neighboring trajectories separate, the divergence is positive and the density decreases. When neighboring trajectories converge, the divergence is negative and the density increases.

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This requires the derivative of $p_t(x_t)$ along a trajectory.

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Both the density function and its input depend on time. Applying the multivariable chain rule gives:

$$\frac{d}{dt}p_t(x_t) = \frac{\partial p_t(x_t)}{\partial t} + \frac{dx_t}{dt} \cdot \nabla p_t(x_t)$$

Substitute the ODE.

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The expanded continuity equation states

$$\frac{\partial p_t(x_t)}{\partial t} + v_t(x_t) \cdot \nabla p_t(x_t) = -p_t(x_t) \nabla \cdot v_t(x_t)$$

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The rate of change of the density $p_t(x_t)$ experienced along the moving trajectory x_t equals negative $p_t(x_t)$ multiplied by the divergence of the velocity field v_t at x_t.

Substitute the ODE.

Positive divergence causes the density along the trajectory to decrease. Negative divergence causes it to increase. Zero divergence preserves the density experienced by the moving sample.

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Dividing by the density produces an especially useful equation for log density.

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For positive density

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The rate of change of the log density log $p_t(x_t)$ along the trajectory x_t equals the negative divergence of the velocity field v_t evaluated at x_t.

The divergence directly determines the instantaneous change in log density.

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A divergence of 0.4 means that the log density decreases at an instantaneous rate of 0.4 along the trajectory. A divergence of negative 0.4 means that the log density increases at an instantaneous rate of 0.4.

Substitute the trajectory-density equation.

$$\boxed{\frac{d}{dt}\log p_t(x_t) = -\nabla \cdot v_t(x_t)}$$

The rate of change of the log density $\log p_t(x_t)$ along the trajectory x_t equals the negative divergence of the velocity field v_t evaluated at x_t.

Integrating from time 0 to time t

$$\log p_t(x_t) = \log p_0(x_0) - \int_0^t \nabla \cdot v_s(x_s)\, ds$$

The log density log $p_t(x_t)$ at time t equals the initial log density log $p_0(x_0)$ minus the velocity divergence $\nabla \cdot v_s(x_s)$ accumulated along the trajectory from time 0 to time t.